IT-SIMPLERA INSTITUTE
Cybersecurity Analyst Internship Program

Week 02 Assignment Report

# COMPUTER NETWORKING FUNDAMENTALS & NETWORK CONFIGURATION

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Week : 02
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# TABLE OF CONTENTS

# Introduction

Computer networking is an important part of today's digital world because it allows computers and other devices to communicate and share information. As someone learning cybersecurity, understanding how networks operate is essential because almost every cyberattack happens over a network.

This assignment has given me the opportunity to explore basic networking concepts by using Windows networking commands, learning the differences between networking devices, comparing IPv4 and IPv6, and studying the OSI Model. Completing these practical exercises helped me understand how devices communicate and how networking problems can be identified and resolved.

# Task 1 – Network Information

### Command Used

ipconfig /all

Using the **ipconfig /all** command, I was able to view the complete network configuration of my computer.

| Network Information | Result |
|---|---|
| IPv4 Address | 10.219.186.195 |
| MAC Address | 50-E0-85-46-26-A2 |
| Default Gateway | 10.219.186.68 |

### IPv4 Address

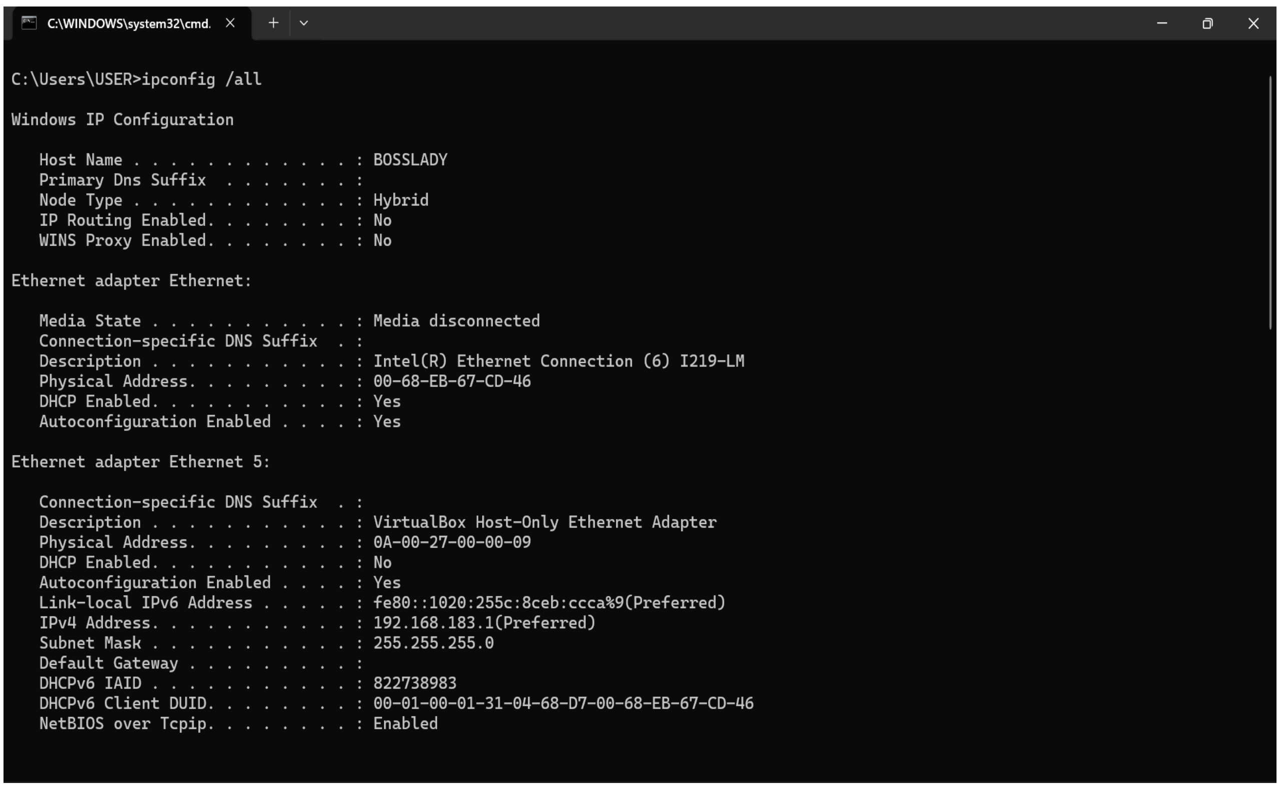
The IPv4 address is the unique address assigned to my computer on the network. It allows my computer to send and receive information from other devices and connect to online services.

### MAC Address

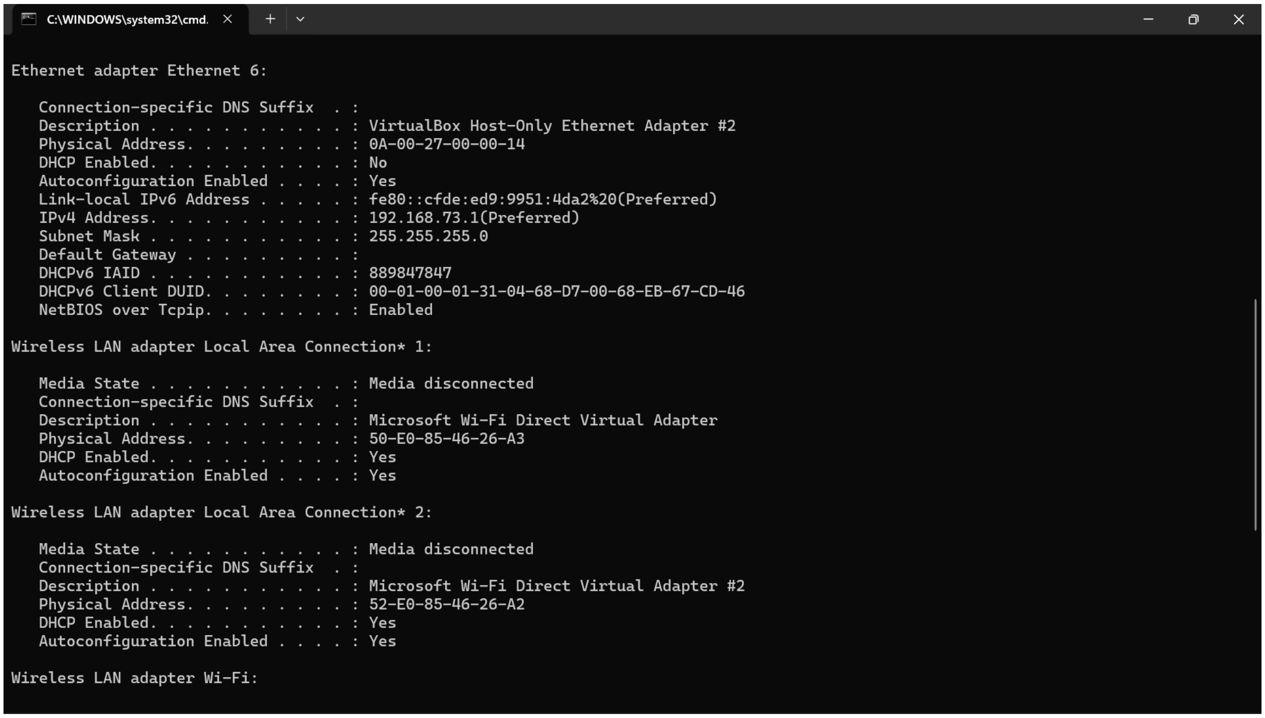
The MAC address is the physical address assigned to my wireless network adapter by the manufacturer. Unlike an IP address, it does not normally change and is used to identify my device on the local network.

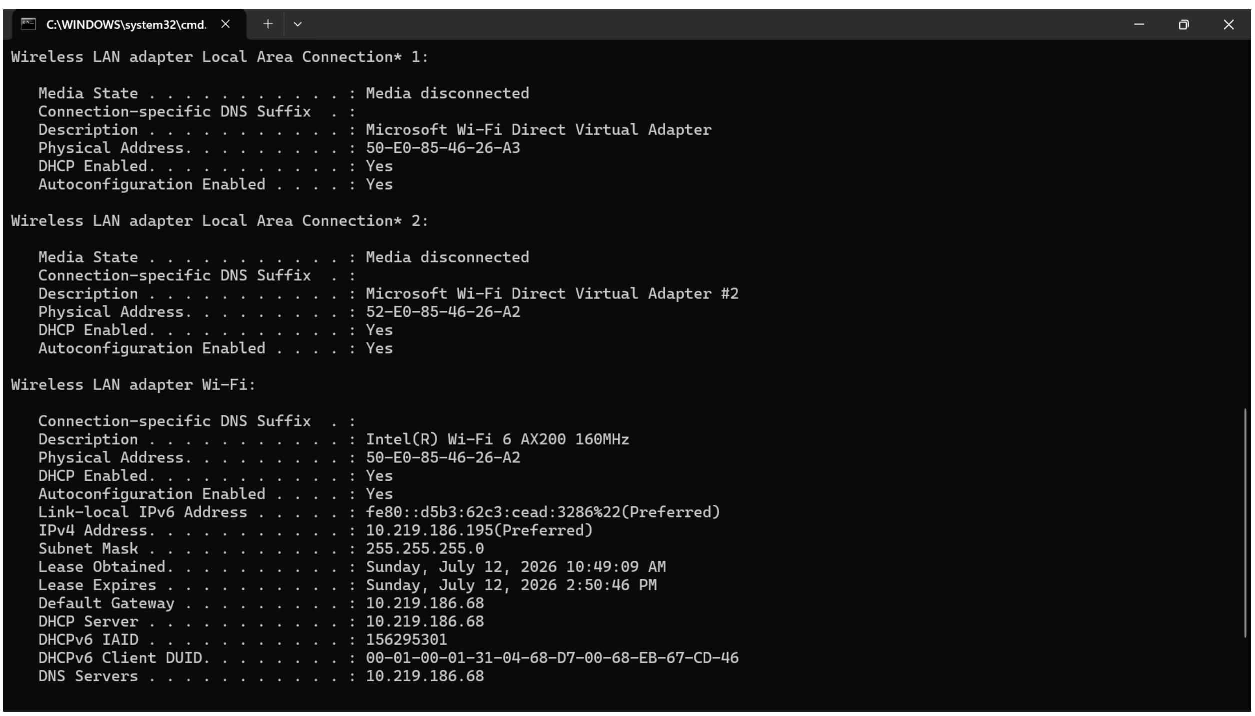
## Default Gateway

The default gateway is the device that connects my local network to other networks, including the internet. Whenever my computer needs to communicate with devices outside my network, the data is first sent to the gateway.



```
C:\WINDOWS\system32\cmd.

C:\Users\USER>ipconfig /all

Windows IP Configuration

   Host Name . . . . . . . . . . . . : BOSSLADY
   Primary Dns Suffix  . . . . . . . :
   Node Type . . . . . . . . . . . . : Hybrid
   IP Routing Enabled. . . . . . . . : No
   WINS Proxy Enabled. . . . . . . . : No

Ethernet adapter Ethernet:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Intel(R) Ethernet Connection (6) I219-LM
   Physical Address. . . . . . . . . : 00-68-EB-67-CD-46
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes

Ethernet adapter Ethernet 5:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : VirtualBox Host-Only Ethernet Adapter
   Physical Address. . . . . . . . . : 0A-00-27-00-00-09
   DHCP Enabled. . . . . . . . . . . : No
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::1020:255c:8ceb:ccca%9(Preferred)
   IPv4 Address. . . . . . . . . . . : 192.168.183.1(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :
   DHCPv6 IAID . . . . . . . . . . . : 822738983
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-31-04-68-D7-00-68-EB-67-CD-46
   NetBIOS over Tcpip. . . . . . . . : Enabled
```


C:\WINDOWS\system32\cmd.
Ethernet adapter Ethernet 6:
Connection-specific DNS Suffix . :
Description . . . . . . . . . . . : VirtualBox Host-Only Ethernet Adapter #2
Physical Address. . . . . . . . . : 0A-00-27-00-00-14
DHCP Enabled. . . . . . . . . . . : No
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::cfde:ed9:9951:4da2%20(Preferred)
IPv4 Address. . . . . . . . . . . : 192.168.73.1(Preferred)
Subnet Mask . . . . . . . . . . . : 255.255.255.0
Default Gateway . . . . . . . . . :
DHCPv6 IAID . . . . . . . . . . . : 889847847
DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-31-04-68-D7-00-68-EB-67-CD-46
NetBIOS over Tcpip. . . . . . . . : Enabled
Wireless LAN adapter Local Area Connection* 1:
Media State . . . . . . . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . . . . . . . : Microsoft Wi-Fi Direct Virtual Adapter
Physical Address. . . . . . . . . : 50-E0-85-46-26-A3
DHCP Enabled. . . . . . . . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Wireless LAN adapter Local Area Connection* 2:
Media State . . . . . . . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . . . . . . . : Microsoft Wi-Fi Direct Virtual Adapter #2
Physical Address. . . . . . . . . : 52-E0-85-46-26-A2
DHCP Enabled. . . . . . . . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Wireless LAN adapter Wi-Fi:


C:\WINDOWS\system32\cmd.
Wireless LAN adapter Local Area Connection* 1:
Media State . . . . . . . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . . . . . . . : Microsoft Wi-Fi Direct Virtual Adapter
Physical Address. . . . . . . . . : 50-E0-85-46-26-A3
DHCP Enabled. . . . . . . . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Wireless LAN adapter Local Area Connection* 2:
Media State . . . . . . . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . . . . . . . : Microsoft Wi-Fi Direct Virtual Adapter #2
Physical Address. . . . . . . . . : 52-E0-85-46-26-A2
DHCP Enabled. . . . . . . . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Wireless LAN adapter Wi-Fi:
Connection-specific DNS Suffix . :
Description . . . . . . . . . . . : Intel(R) Wi-Fi 6 AX200 160MHz
Physical Address. . . . . . . . . : 50-E0-85-46-26-A2
DHCP Enabled. . . . . . . . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::d5b3:62c3:cead:3286%22(Preferred)
IPv4 Address. . . . . . . . . . . : 10.219.186.195(Preferred)
Subnet Mask . . . . . . . . . . . : 255.255.255.0
Lease Obtained. . . . . . . . . . : Sunday, July 12, 2026 10:49:09 AM
Lease Expires . . . . . . . . . . : Sunday, July 12, 2026 2:50:46 PM
Default Gateway . . . . . . . . . : 10.219.186.68
DHCP Server . . . . . . . . . . . : 10.219.186.68
DHCPv6 IAID . . . . . . . . . . . : 156295301
DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-31-04-68-D7-00-68-EB-67-CD-46
DNS Servers . . . . . . . . . . . : 10.219.186.68

# Task 2 – Basic Networking Commands

## 1. ipconfig /all

This command displays detailed information about every network adapter installed on my computer. It provides useful details such as the IP address, MAC address, DNS server, subnet mask, and default gateway. It is one of the first commands used when troubleshooting network problems.

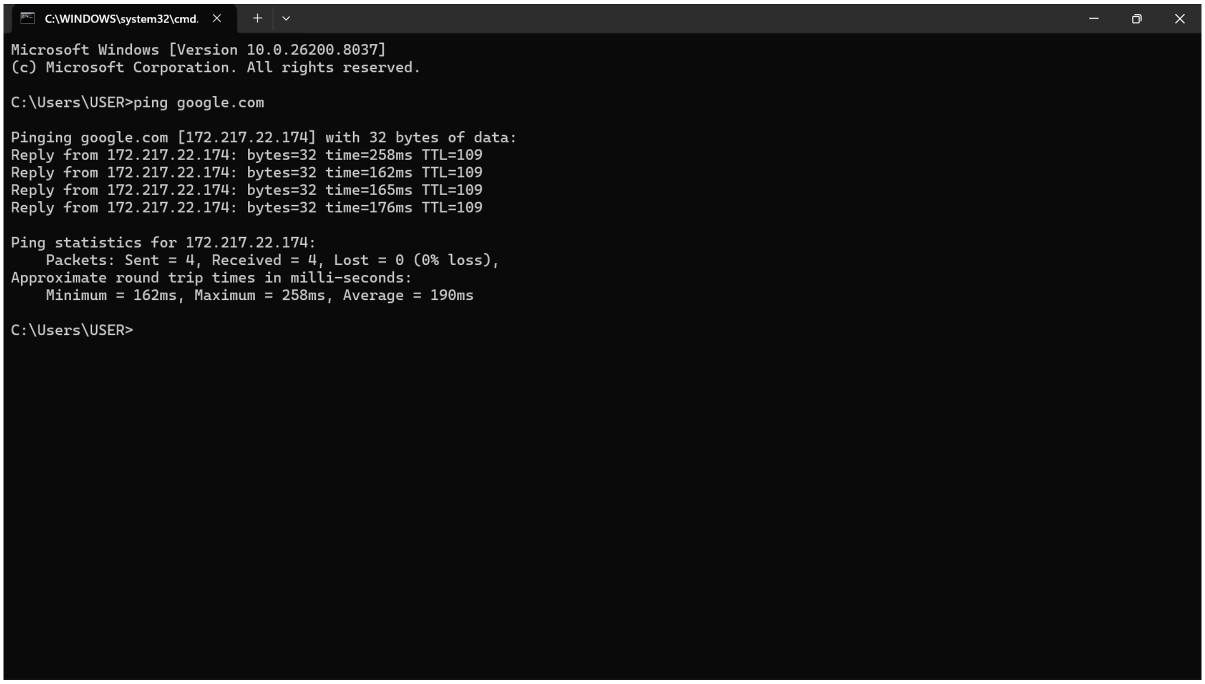
## 2. ping google.com

The **ping** command checks whether another computer or website can be reached over the network. It also measures how long it takes for data to travel between my computer and the destination.

### My Result

- Packets Sent: **4**
- Packets Received: **4**
- Packet Loss: **0%**
- Average Response Time: **190 ms**

Since all four packets were received successfully and there was no packet loss, the result shows that my internet connection was working properly during the test.

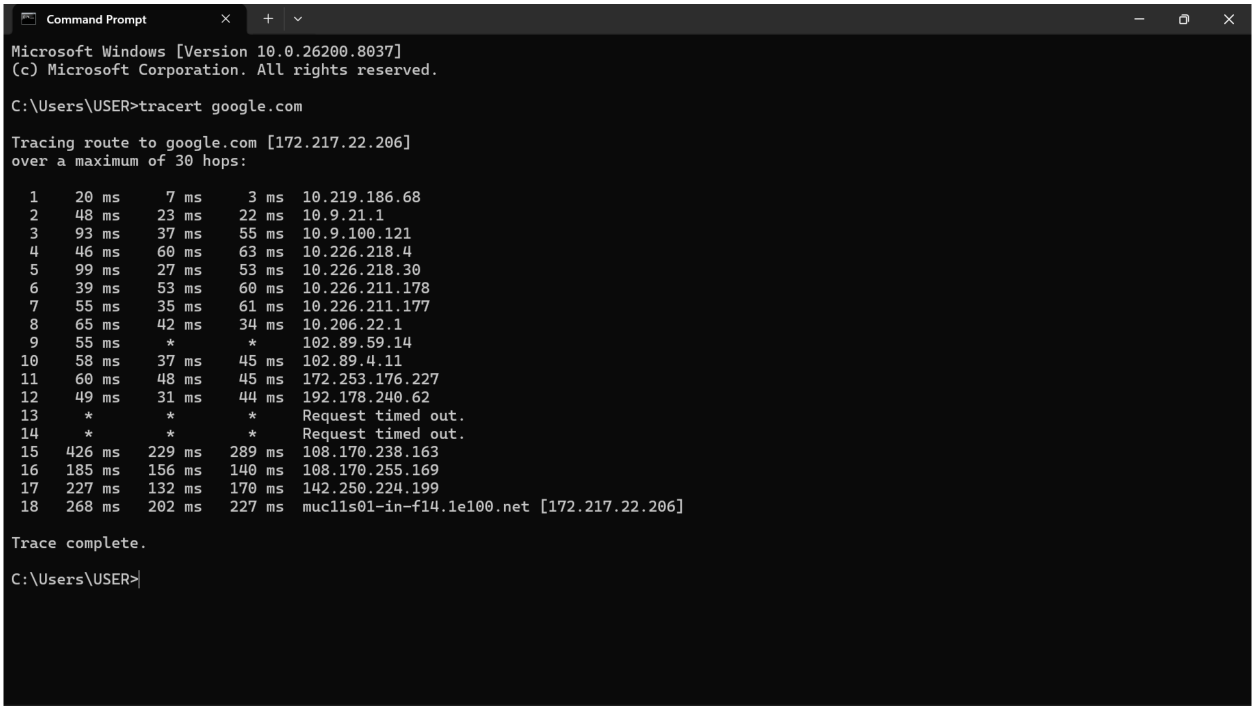


```
C:\WINDOWS\system32\cmd.
Microsoft Windows [Version 10.0.26200.8037]
(c) Microsoft Corporation. All rights reserved.

C:\Users\USER>ping google.com

Pinging google.com [172.217.22.174] with 32 bytes of data:
Reply from 172.217.22.174: bytes=32 time=258ms TTL=109
Reply from 172.217.22.174: bytes=32 time=162ms TTL=109
Reply from 172.217.22.174: bytes=32 time=165ms TTL=109
Reply from 172.217.22.174: bytes=32 time=176ms TTL=109

Ping statistics for 172.217.22.174:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 162ms, Maximum = 258ms, Average = 190ms

C:\Users\USER>
```

## 3. tracert google.com

The **tracert** command shows the path that network packets take before reaching their destination. Every stop along the journey is called a **hop**.

During my test, the packets successfully reached Google's server after passing through several routers. I also noticed that a few hops displayed **"Request Timed Out."** This does not necessarily indicate a problem because some routers are configured not to respond to traceroute requests for security reasons.



```
Microsoft Windows [Version 10.0.26200.8037]
(c) Microsoft Corporation. All rights reserved.

C:\Users\USER>tracert google.com

Tracing route to google.com [172.217.22.206]
over a maximum of 30 hops:

  1    20 ms     7 ms     3 ms  10.219.186.68
  2    48 ms    23 ms    22 ms  10.9.21.1
  3    93 ms    37 ms    55 ms  10.9.100.121
  4    46 ms    60 ms    63 ms  10.226.218.4
  5    99 ms    27 ms    53 ms  10.226.218.30
  6    39 ms    53 ms    60 ms  10.226.211.178
  7    55 ms    35 ms    61 ms  10.226.211.177
  8    65 ms    42 ms    34 ms  10.206.22.1
  9    55 ms     *        *     102.89.59.14
 10    58 ms    37 ms    45 ms  102.89.4.11
 11    60 ms    48 ms    45 ms  172.253.176.227
 12    49 ms    31 ms    44 ms  192.178.240.62
 13     *        *        *     Request timed out.
 14     *        *        *     Request timed out.
 15   426 ms   229 ms   289 ms  108.170.238.163
 16   185 ms   156 ms   140 ms  108.170.255.169
 17   227 ms   132 ms   170 ms  142.250.224.199
 18   268 ms   202 ms   227 ms  muc11s01-in-f14.1e100.net [172.217.22.206]

Trace complete.

C:\Users\USER>
```

## 4. nslookup google.com

The **nslookup** command is used to check how a domain name is translated into an IP address by the Domain Name System (DNS).

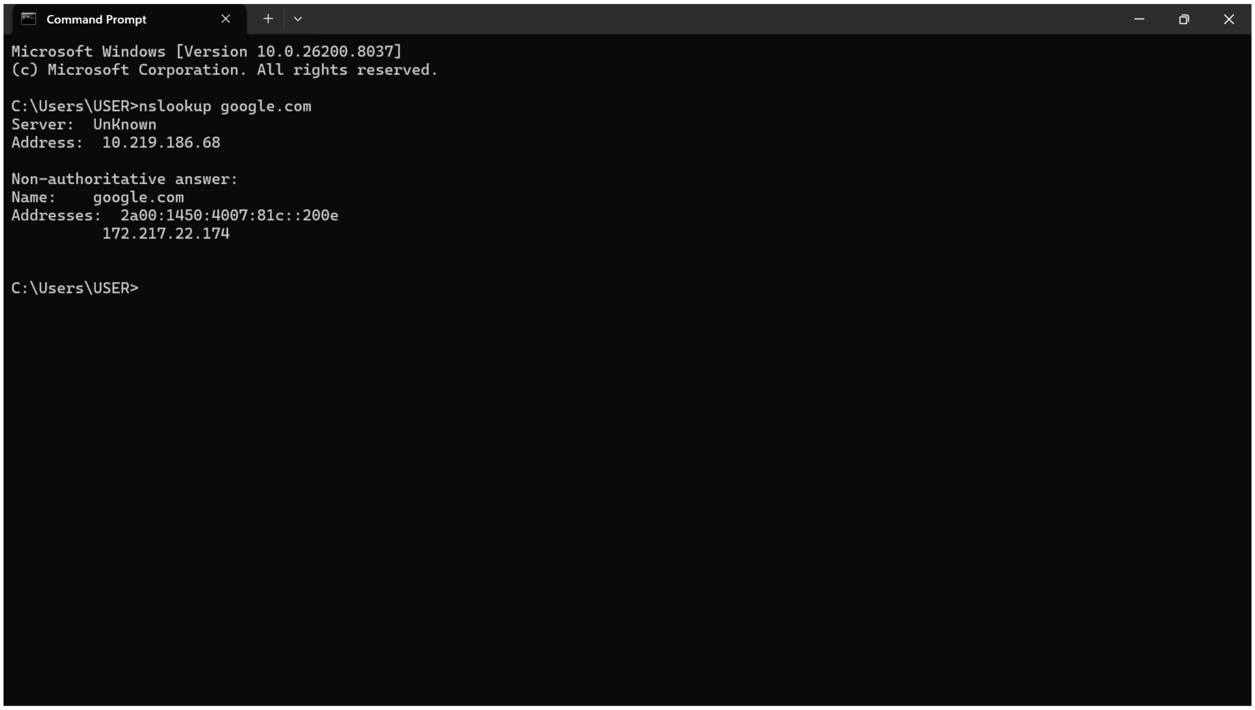
### My Result

DNS Server:
**10.219.186.68**

IPv4 Address:
**142.251.39.206**

IPv6 Address:
**2a00:1450:4007:81c::200e**

The results confirm that my DNS server successfully translated **google.com** into both its IPv4 and IPv6 addresses.




## 5. arp -a

The **arp -a** command displays the Address Resolution Protocol (ARP) table stored on my computer. The table contains IP addresses and their corresponding MAC addresses for devices that my computer has recently communicated with on the local network.

This command is useful when troubleshooting network communication because it helps identify devices connected to the same network.

Command Prompt

```
Microsoft Windows [Version 10.0.26200.8037]
(c) Microsoft Corporation. All rights reserved.

C:\Users\USER>arp -a

Interface: 192.168.183.1 --- 0x9
  Internet Address      Physical Address      Type
  192.168.183.255       ff-ff-ff-ff-ff-ff     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  239.255.255.250       01-00-5e-7f-ff-fa     static

Interface: 192.168.73.1 --- 0x14
  Internet Address      Physical Address      Type
  192.168.73.255        ff-ff-ff-ff-ff-ff     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  239.255.255.250       01-00-5e-7f-ff-fa     static

Interface: 10.219.186.195 --- 0x16
  Internet Address      Physical Address      Type
  10.219.186.68         76-af-21-1a-1f-3a     dynamic
  10.219.186.255        ff-ff-ff-ff-ff-ff     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  239.255.255.250       01-00-5e-7f-ff-fa     static
  255.255.255.255       ff-ff-ff-ff-ff-ff     static

C:\Users\USER>
```

# Task 3 – Router vs Switch

## Router

A **router** is a networking device that connects different networks together, such as a home network and the internet. It uses IP addresses to send data to the correct destination and allows multiple devices to share the same internet connection.

## Switch

A **switch** is a networking device that connects multiple devices within the same local network (LAN). It uses MAC addresses to forward data only to the intended device, making communication within the network faster and more efficient.

| **Router** | **Switch** |
|---|---|
| Connects different networks together. | Connects devices within the same network. |
| Operates at Layer 3 of the OSI Model. | Operates at Layer 2 of the OSI Model. |
| Uses IP addresses to forward data. | Uses MAC addresses to forward data. |
| Connects local networks to the internet. | Connects computers, printers, and other devices within a LAN. |
| Usually provides additional services such as DHCP and NAT. | Mainly forwards data between connected devices. |
| Slightly slower because it performs routing decisions. | Generally faster because it switches data directly between devices. |

# Task 4 – IPv4 vs IPv6

## IPv4

IPv4 (Internet Protocol Version 4) is the most widely used version of the Internet Protocol. It uses **32-bit addresses** (e.g., **192.168.1.1**) to identify devices on a network. Although it is reliable and widely supported, the limited number of available addresses has led to the development of IPv6.

## IPv6

IPv6 (Internet Protocol Version 6) is the newer version of the Internet Protocol. It uses **128-bit addresses** (e.g., **2001:db8::1**) to provide a much larger address space. It also offers improved security, better performance, and supports the growing number of devices connected to the internet.

| IPv4 | IPv6 |
|---|---|
| Uses a 32-bit addressing system. | Uses a 128-bit addressing system. |
| Supports about 4.3 billion unique addresses | Supports an extremely large number of addresses. |
| Written in decimal format. | Written in hexadecimal format. |
| Requires NAT in many networks. | Designed to eliminate the need for NAT. |
| Security features are optional. | Built with stronger security support through IPSec. |
| Smaller address space. | Much larger address space for future growth. |

# Task 5 – OSI Model

## OSI Model

The **OSI (Open Systems Interconnection) Model** is a framework that explains how data moves from one device to another over a network. It divides network communication into **seven layers**, with each layer performing a specific function. The model helps network engineers and cybersecurity professionals understand, troubleshoot, and secure network communications more effectively.

| **Layer** | **Primary Function** | **Protocol** | **Device** |
|---|---|---|---|
| 7. Application | Provides services directly to users and applications. | HTTP | Computer |
| 6. Presentation | Formats, encrypts, and compresses data. | SSL/TLS | Gateway |
| 5. Session | Creates, manages, and terminates communication sessions. | NetBIOS | Gateway |
| 4. Transport | Ensures reliable delivery of data between devices. | TCP | Firewall |
| 3. Network | Determines the best route for data to travel. | IP | Router |
| 2. Data Link | Uses MAC addresses to transfer data within the local network. | Ethernet | Switch |
| 1. Physical | Transmits raw bits through cables or wireless signals. | Ethernet | Network Cable/hub |

# Challenges

At the beginning of this assignment, some of the networking commands looked confusing, especially **tracert**, **arp**, and **nslookup**, because I had never used them before. After carrying out the commands and studying their outputs, I developed a better understanding of what each one does and how they can help diagnose network problems.

Another challenge was understanding the relationship between the OSI layers and real networking devices. Learning the purpose of each layer made it easier to understand how data moves from one computer to another.

# Conclusion

This assignment strengthened my understanding of basic computer networking concepts and gave me practical experience using important Windows networking commands. I learned how to identify my computer's network configuration, test connectivity, trace network paths, resolve domain names, and view the ARP table.

I also gained a clearer understanding of networking devices, IP addressing, and the OSI Model. These concepts provide a solid foundation for learning more advanced networking and cybersecurity topics in the future.

## References


- Microsoft Learn – Windows Networking Documentation
- Cisco Networking Academy (NetAcad)
- CompTIA Network+ Study Guide
- Coursera.org article